

Pokhara University
Faculty of Science and Technology

Course Code: CMP 559 (3 Credits)
Course Title: Theory of Computation (3-0-0)
Nature of the course: Theory
Level: Master

Full marks: 100
Pass marks: 60
Total Lectures: 45 hrs
Program: MSc CS

1. Course Description

This course provides an introduction to the fundamentals of computability and formal languages, including symbols, alphabets, and strings. Students will model problems using finite automata, regular expressions, and context-free grammars. They will explore Turing machines, undecidable problems, and the limits of computation. The course covers feasible versus intractable problems, NP-completeness, and problem reductions. Students will also study computational complexity and asymptotic analysis. By the end, they will understand the foundations of efficient and parallel computation models.

2. General Objectives

The course is designed with the following general objectives:

- To provide the students with essential knowledge of computability concepts, computational boundaries, and the theoretical foundations of computer science.
- To enable the students to design and analyze computational models such as automata, grammars, and Turing machines.
- To acquaint the students with principles of computational hardness, NP-completeness, efficient computation strategies, and parallel processing models.

3. Contents in Detail

Specific Objectives	Contents
Explain the capabilities and limitations of computability	1. Introduction (3 hrs) 1.1. Some computing puzzles 1.2. Understanding computability
Explain formal languages, operations, problem types, and their computational representation.	2. Languages (6 hrs) 2.1. Basic Concepts - Symbols, Alphabets, Strings, Alphabet Encoding 2.2. Languages and Operations 2.3. Problem Representation 2.4. Types of Problems - Representing Graphs, Spanning Tree, Decision Problem, Function Problem, Search Problem 2.5. Casting Problems into Languages
Describe regular languages, finite automata, regular expressions, and pumping lemma with their applications	3. Finite State Machines and Regular Languages (8 hrs) 3.1 Regular Languages 3.2 Regular Expressions 3.2.1 Applications of Regular Expressions 3.3 Finite Automata 3.3.1 Deterministic Finite Automata (DFA) 3.3.2 Non-Deterministic Finite Automata (NFA)

	3.3.3 Equivalence of DFA and NFA 3.4 Properties of Finite State Languages 3.5 Pumping Lemma for Regular Languages 3.5.1 Applications of Pumping Lemma
Describe grammars, pushdown automata, and parsing techniques with parse trees.	4. Pushdown Automata and Context Free Languages (6 hrs) 4.1. Grammars 4.1.1 Regular Grammars 4.1.2 Context-Free Grammars (CFGs) 4.2. Pushdown Automata (PDA) 4.3. Parsing Techniques 4.3.1 Parse Trees 4.3.2 Attribution of Parse Tree
Analyze Turing machines and undecidability in computability theory.	5. Turing Machine and Computability (6 hrs) 5.1. Concept of Turing Machines 5.2. Basic Machine Notions 5.3. Church-Turing Thesis 5.4. Undecidable Languages
Describe feasible and intractable problems, polynomial-time complexity, problem transformation techniques, and NP-completeness.	6. Feasible and Intractable Problems (8 hrs) 6.1 Polynomial-Time 6.2 P-Completeness Theory and Problems 6.3 Reducibility 6.4 NP-Completeness Theory and Problems
Examine asymptotic complexity, time/space tradeoffs, and the NC class in computational complexity	7. Computational Complexity Theory (8 hrs) 7.1 Overview of Computational Complexity 7.2 Asymptotic Notation 7.3 Time and Space Complexity 7.4 Complexity Class NC

4. Method of Instruction

- Lectures and conceptual explanations
- Assist students to gain practical skills in computational theory
- Research assignments
- Classroom discussions

5. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, internal assessment, project and research works etc. The internal evaluation scheme for this course is as follows (but not limited to):

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Attendance & Class Participation	10%	60	Semester-End examination	40
Assignments	20%			

Internal Assessment	40%			
Research/Project/Case Study Work	30%			
Total Internal		60	Total External	40
Full Marks: 60 + 40 = 100				

Students' Responsibilities

Each student must secure at least 60% marks separately in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

6. Reference Books

1. Lewis H.R. and C.H. Papadimitriou, Elements of the theory of computation, 1998, Pearson Education.
2. Cohesn Daniel, I.A., Introduction to computer theory, 2E, 2000, John Woley and sons (Asia) Pvt. Ltd. Singapore.
3. Mishra, K.L.P. and Chandrasekhar, N., Theory of Computer Science, PHI, New Delhi, 1998.
4. Hopcroft, J.E. and Ulman, J.D., Introduction to Automata Theory, Languages and Computation, Narosa Publishing House, New Delhi.
5. Michael Sipser, Introduction to the Theory of Computation, Cengage Learning.